DOCUMENT 03 — FLOW AUTOMATION

Flow Builder Implementations

Smart Real Estate System

Salesforce CRM and Agentforce Implementation

Prepared by	Rayian Hussain Harboosh
Document version	1.0
Evidence basis	Active Flow metadata and Salesforce Builder screenshots
Date	August 28, 2026

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Scope note: This document records implementation structure and configuration evidence. Test execution screenshots and Actual Results remain in the separate testing deliverable.

1. Flow Portfolio and Evidence Basis

Implementation extracted from the supplied API v67.0 Flow metadata and full-canvas screenshots.

The solution contains four active Autolaunched Flows mapped directly to Agentforce actions and two active supporting application Flows. All four Agentforce Flows execute in System Mode Without Sharing and therefore require tightly scoped inputs, queries, and actions.

Flow	Type / status	Role
Update Lead Preferences	Autolaunched / Active	Finds a Lead by email and updates property preferences.
Check Agent Availability	Autolaunched / Active	Detects overlapping non-cancelled showings for an agent.
Create Showing From Agent	Autolaunched / Active	Validates property/time, checks conflicts, and creates a Showing.
Search Available Properties	Autolaunched / Active	Retrieves available Property records for Agentforce.
Lead Qualification Before Save	Before-save record-triggered / Active	Calculates Lead score and qualification status.
Schedule Property Showing	Screen Flow / Active	Interactive internal showing-scheduling experience.

> **Evidence interpretation:** The first four Flows are directly exposed through Agentforce Actions. The last two are supporting Salesforce automations and are not presented as Agentforce Actions.

1.1 Documentation Conventions

- Inputs and outputs are listed from metadata, including variables currently declared but not consumed or assigned.
- Expected business failures are separated from unexpected platform exceptions.
- No red Fault Connectors are present in the supplied Flow canvases.

2. Agentforce Action Mapping

Agentforce topic	Action label / API name	Referenced Flow
Lead Qualification	Update Lead Preferences / Lead_Qual	Update Lead Preferences
Agent Availability	Check Agent Availability / Check_Agent_Availability	Check Agent Availability
Schedule Showing	Create Showing / Create_Showing	Create Showing From Agent
Property Details Inquiry	Get Property Details / Get_Property_Details	Search Available Properties

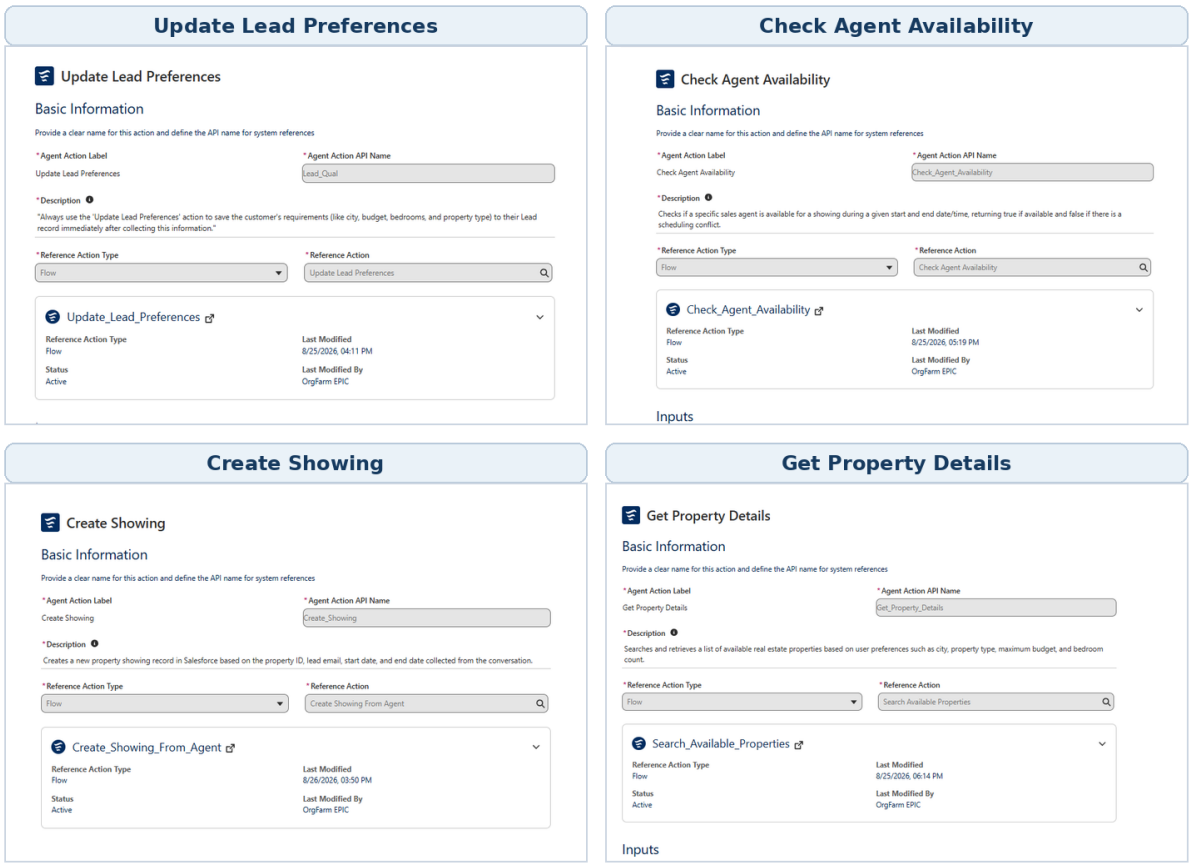


Figure 1. Agentforce action definitions and their referenced Autolaunched Flows.

> **Control principle:** Agentforce chooses a governed action; the action invokes a specific Flow. The conversational model does not receive unrestricted database operations.

3. Update Lead Preferences — Full Canvas

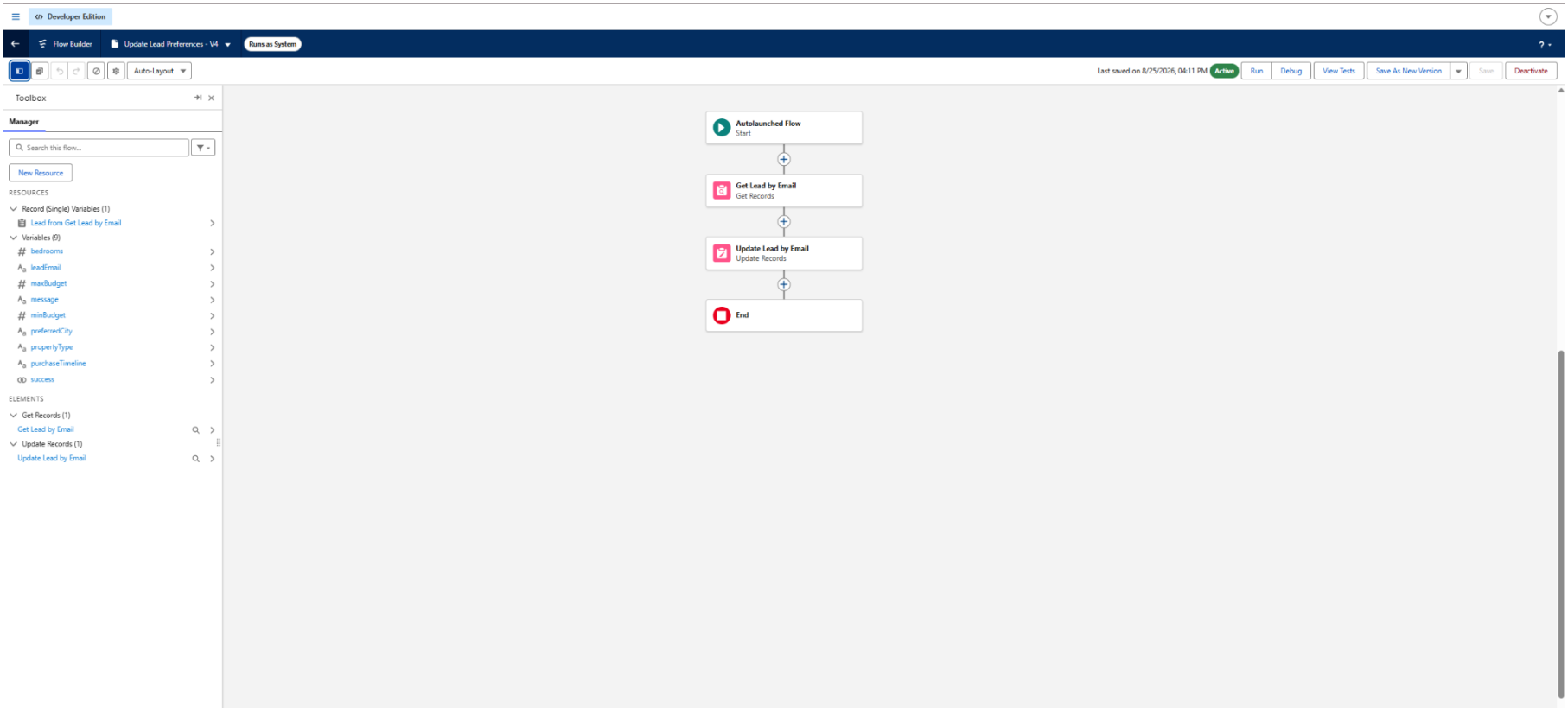


Figure 2. Update Lead Preferences V4 — active Autolaunched Flow canvas.

> **Canvas sequence:** Start → Get Lead by Email → Update Lead by Email → End

3.1 Update Lead Preferences — Implementation Detail

Purpose: Locate one Lead by email and save the customer's structured real-estate preferences.

Direction	Variable	Type / use
Input	leadEmail	String; used to locate the Lead
Input	bedrooms	Number(2); mapped to Bedrooms Required
Input	minBudget / maxBudget	Number(2); mapped to budget fields
Input	preferredCity	String; mapped to Preferred City
Input	propertyType	String; mapped to Preferred Property
Input	purchaseTimeline	String; mapped to Purchase Timeline
Output	success / message	Boolean and String; declared but not assigned in current metadata

3.2 Record Operations

Element	Operation	Configuration
Get Lead by Email	Get Records	Lead where Email = leadEmail; first record only
Update Lead by Email	Update Records	Updates the retrieved Lead by Id

Implementation note: The Flow declares success and message outputs, but the supplied metadata contains no Assignments that populate them. The record update may still complete, but the Agentforce response contract is not explicit.

Error behavior: There is no Decision for a missing Lead and no explicit Fault Connector. A stronger version would add a Lead-found Decision plus success/message Assignments and technical fault paths.

4. Check Agent Availability — Full Canvas

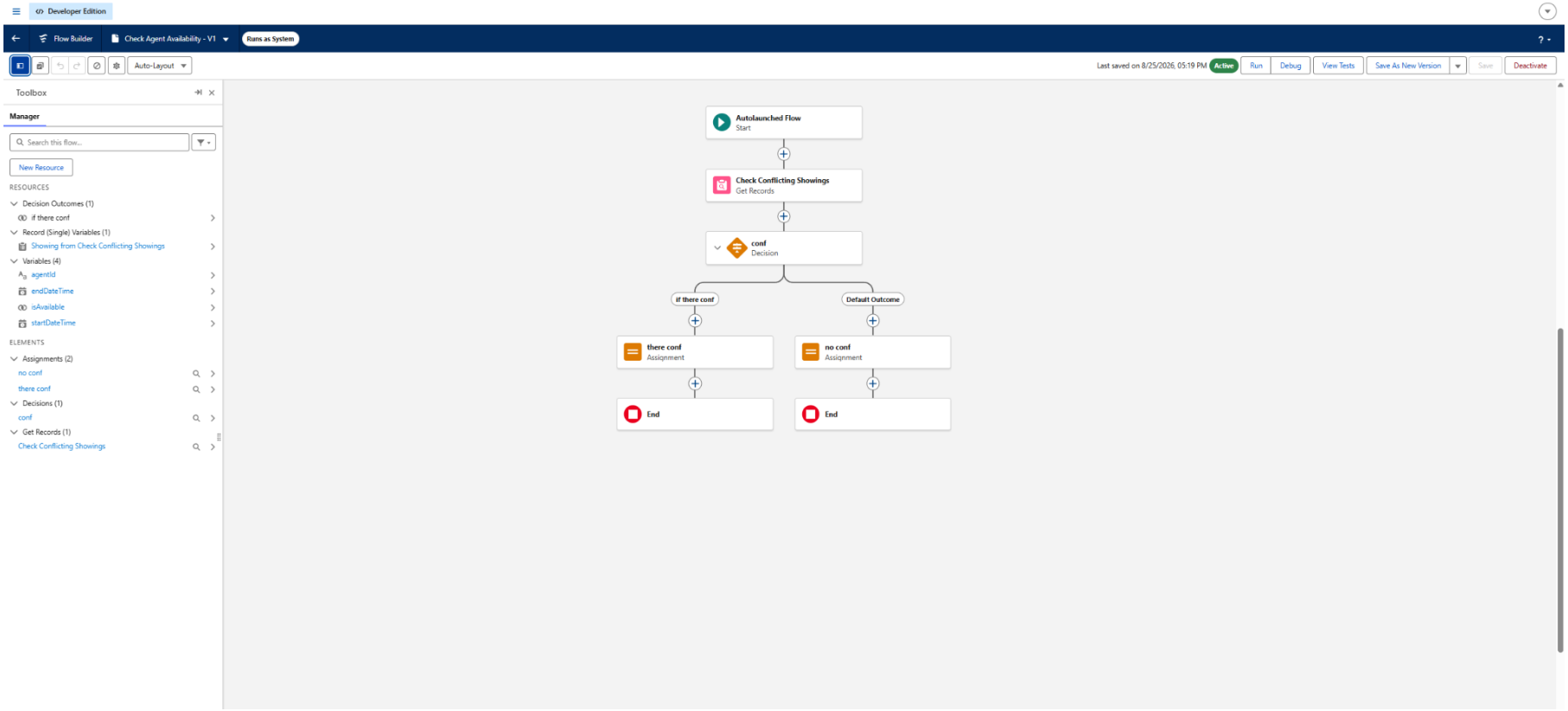


Figure 3. Check Agent Availability V1 — active Autolaunched Flow canvas.

Canvas sequence: Start → Check Conflicting Showings → Decision → isAvailable assignment → End

4.1 Check Agent Availability — Implementation Detail

Purpose: Return whether a selected sales agent is free for a requested start/end interval.

Direction	Variable	Type
Input	agentId	String
Input	startDateTime	DateTime
Input	endDateTime	DateTime
Output	isAvailable	Boolean; default true

4.2 Conflict Query and Decision

Stage	Logic
Get Records	Find the first Showing where Agent = agentId, Start < requested end, End > requested start, and Status ≠ Cancelled.
Conflict outcome	If a Showing is found, assign isAvailable = false.
Default outcome	If no matching Showing is found, assign isAvailable = true.

Controlled negative outcome: A scheduling conflict is expected business logic, not a Flow crash. The Decision returns false to Agentforce so the conversation can offer another time.

Technical exception note: The Flow has no explicit Fault Connector for failures such as query/permission/platform errors. Those exceptions are distinct from the no-availability business outcome.

5. Create Showing From Agent — Full Canvas

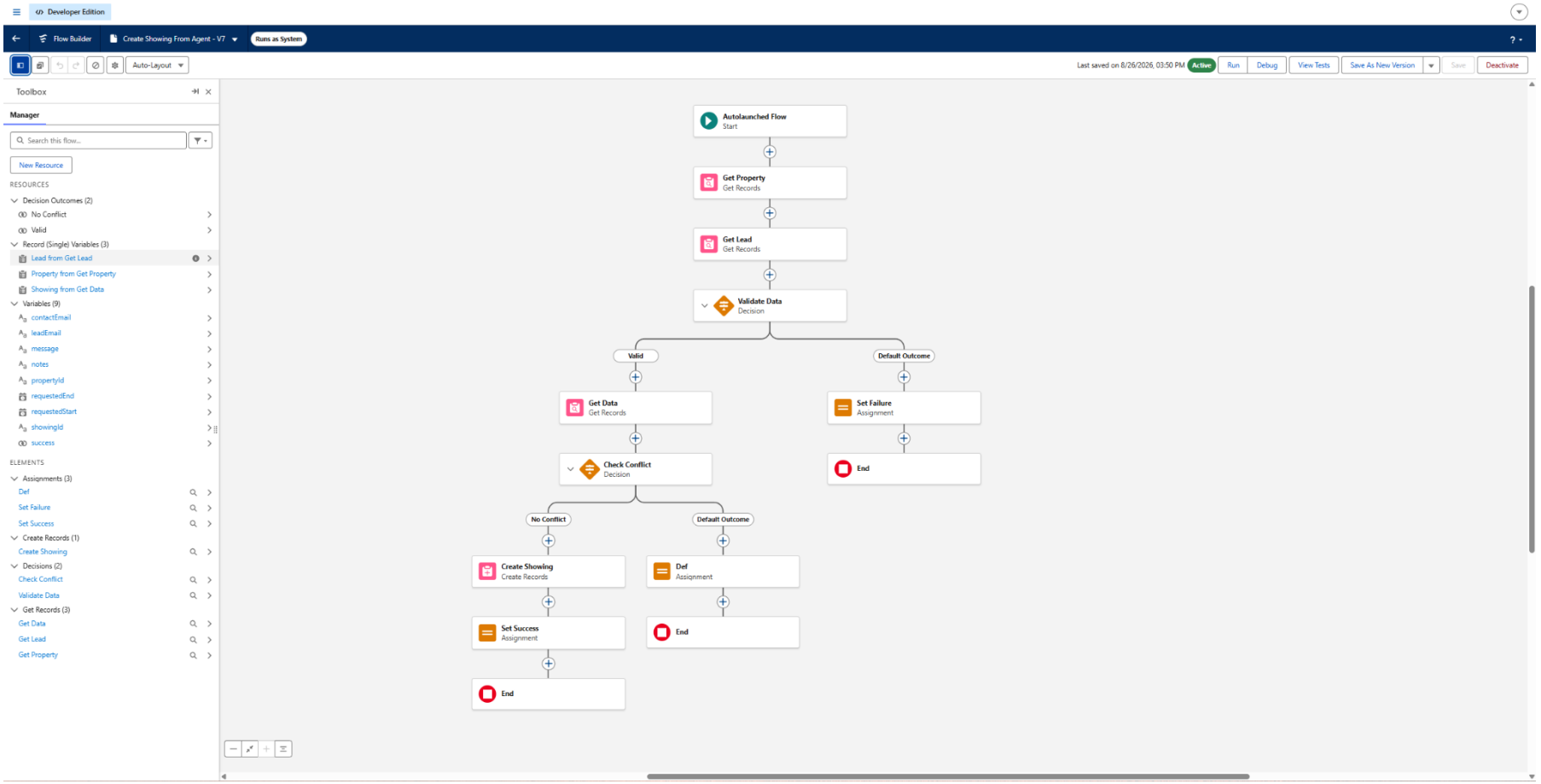


Figure 4. Create Showing From Agent V7 — active Autolaunched Flow canvas.

5.1 Create Showing — Inputs, Lookups, and Validation

Purpose: Create a requested property showing only after the property, Lead, and requested interval pass validation and conflict checks.

Direction	Variable	Type / status
Input	propertyId	String; used
Input	leadEmail	String; used
Input	requestedStart / requestedEnd	DateTime; used
Input	contactEmail / notes	String; declared but not used in current metadata
Output	success	Boolean
Output	message	String
Output	showingId	String; created Showing Id

5.2 Validation Decision

Check	Required condition
Property	Property Id is present and Property Status = Available
Time	requestedStart > current date/time
Chronology	requestedEnd > requestedStart

Implementation observation: The Flow retrieves the Lead by email but the validation Decision does not explicitly test that the Lead Id is non-null. A Lead-found check would make the failure message more precise.

5.3 Create Showing — Conflict, Create, and Outcomes

5.3.1 Conflict Detection

The Flow searches for an existing non-cancelled Showing for the same Property where the stored interval overlaps the requested interval. A null result follows the No Conflict branch; a found record follows the conflict branch.

5.3.2 Create Records

Showing field	Mapped value
Property__c	Retrieved Property Id
Lead__c	Retrieved Lead Id
Start_Date_Time__c	requestedStart
End_Date_Time__c	requestedEnd
Status__c	Requested
Created record Id	showingId output

5.3.3 Business Outcomes

Outcome	success	message
Created	true	Showing scheduled successfully.
Invalid/unavailable	false	The selected time is unavailable or the property is not available.
Conflict	false	The selected time is already booked. Please choose another time.

> **Error-path design:** Expected business failures terminate cleanly with success=false and a user-facing message. No explicit Fault Connectors capture unexpected failures from Get Records or Create Records.

6. Search Available Properties — Full Canvas

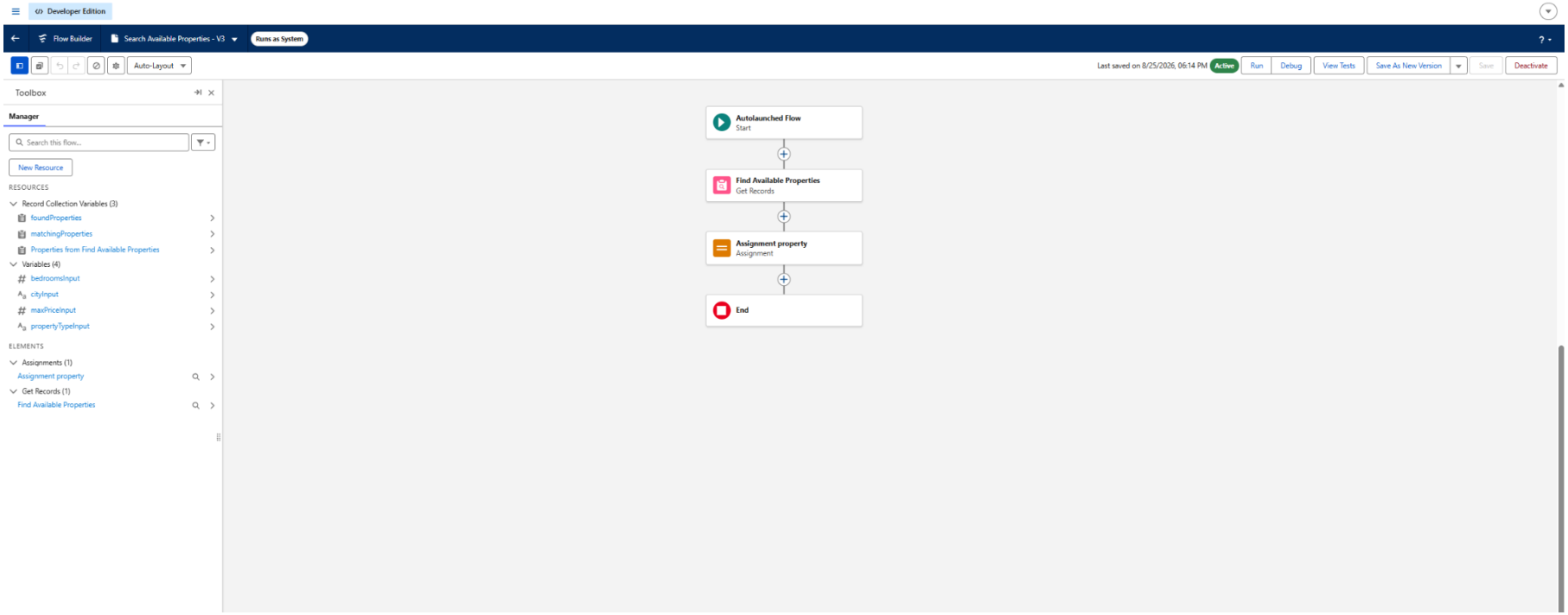


Figure 5. Search Available Properties V3 — active Autolaunched Flow canvas.

Canvas sequence: Start → Find Available Properties → Assignment property → End

6.1 Search Available Properties — Implementation Detail

Purpose: Return available Property records to the Property Details Inquiry topic through the Get Property Details action.

Direction	Variable	Type / current behavior
Input	cityInput	String; declared, not used in current query
Input	maxPriceInput	Number(0); declared, not used in current query
Input	bedroomsInput	Number(0); declared, not used in current query
Input	propertyTypeInput	String; declared, not used in current query
Output	matchingProperties	Property collection; assigned from Get Records
Output	foundProperties	Property collection; declared but not assigned

6.2 Query and Assignment

Element	Current configuration
Find Available Properties	Get all Property records where Status__c = Available.
Assignment property	matchingProperties receives the returned collection.

Material implementation note: Although the action description and inputs support city, budget, bedrooms, and property type, the current metadata filters only by availability status. The Flow should add optional filters or documented branching before claiming filtered search behavior.

No-results behavior: There is no explicit Decision for an empty collection and no success/message output. Agentforce must infer no results from an empty returned collection.

7. Error-Handling Architecture

The implementation uses controlled decision paths for expected business conditions. This is appropriate for conflicts, unavailable properties, invalid intervals, or missing matches because these outcomes should be communicated to the agent rather than treated as Flow crashes.

Flow	Expected business handling	Unexpected technical handling
Update Lead Preferences	No explicit missing-Lead branch; outputs declared but unassigned	No Fault Connectors
Check Agent Availability	Decision returns isAvailable true/false	No Fault Connectors
Create Showing	Decisions return success=false and descriptive messages	No Fault Connectors
Search Properties	Empty collection can represent no match; no explicit branch	No Fault Connectors

7.1 Correct Interpretation

Business-error handling: Decisions and Assignments prevent expected negative outcomes from being misrepresented as successful transactions. Create Showing provides the strongest current example through explicit success and message outputs.

Residual technical risk: Decision logic does not replace Fault Connectors for unexpected platform failures such as unhandled DML errors, query failures, or permission exceptions. Production hardening should add fault paths that set safe outputs and avoid exposing internal details.

7.2 Recommended Fault Response Contract

- Set success = false and a concise user-safe message.
- Log diagnostic context for administrators without exposing sensitive data to the conversation.
- Avoid claiming that a record was created or updated when a DML element faults.

8. Supporting Application Flows

8.1 Lead Qualification Before Save

Type: Active before-save record-triggered Flow on Lead for create and update events.

The Flow calculates a qualification score from contact availability and captured preferences, then assigns a status: Qualified at 70 or more, Warm at 50–69, and Nurture below 50.

Criterion	Points
Email or Phone present	20
Maximum Budget present	20
Preferred City present	15
Preferred Property Type present	15
Bedrooms Required present	10
Purchase Timeline present	20

> **Agentforce relationship:** This Flow is not an Agent Action. It runs automatically when Lead data is saved, including updates initiated by the Update Lead Preferences action.

8.2 Schedule Property Showing

Type: Active Screen Flow with recordId input.

The interactive Flow retrieves a Property, collects the requested interval and contact details, checks property status and existing showings, and either creates a Requested Showing or displays a conflict outcome. It supports internal UI use and is separate from the Agentforce Create Showing action.

9. Security, Deployment, and Test Traceability

9.1 Execution Security

The four Agentforce-mapped Flows run in System Mode Without Sharing. This avoids depending on the conversational user's record-sharing context, but it also elevates the importance of strict input validation, narrow queries, minimal record mutation, action access control, and field-level authorization outside the prompt text.

> **Security boundary:** Agent instructions are behavioral controls, not a substitute for Salesforce permissions, sharing, field-level security, action access, or Flow-level authorization.

9.2 Deployment and Versioning

- All documented Flows are Active and use API version 67.0.
- Deploy metadata and dependencies through the authorized Salesforce deployment workflow.
- Activate Agentforce Actions only after the referenced Flow versions and permissions are verified.
- Retain prior Flow versions for controlled rollback; do not edit an active production version in place.

9.3 Test Traceability

Implementation area	Required verification
Lead preferences	Existing Lead update, missing Lead, invalid budgets, and output contract
Availability	No conflict, overlap conflict, cancelled showing, and boundary times
Showing creation	Valid create, unavailable property, past time, invalid chronology, conflict, missing Lead, and DML fault
Property search	Availability, optional filters, no results, large result set, and field visibility

> **Evidence separation:** Actual Results and execution screenshots belong in the test document. This implementation document defines what must be verified and how it maps to each Flow.

10. Conclusion and Evidence Index

The Smart Real Estate System uses four active Autolaunched Flows to expose controlled CRM operations to Agentforce and two supporting Flows for automatic qualification and interactive scheduling. The strongest implementation is Create Showing From Agent, which combines validation, conflict checking, record creation, and explicit business outcomes. The principal hardening priorities are completing action output contracts, applying the declared property-search filters, validating missing-record conditions, and adding Fault Connectors for unexpected technical exceptions.

> **Overall assessment:** The architecture is coherent and demonstrable. It should be described as controlled business-error handling with remaining technical fault-hardening work—not as complete exception handling.

10.1 Evidence Index

Evidence	Used for
FLOW1.png	Update Lead Preferences full canvas
FLOW2.png	Check Agent Availability full canvas
FLOW3.png	Create Showing From Agent full canvas
FLOW4.png	Search Available Properties full canvas
FLOW5–FLOW8.png	Agentforce Action definitions
Flows_Metadata_Final package	Five original Flow metadata files
Update_Lead_Preferences.flow-meta.xml	Final missing Flow metadata supplied separately

10.2 Submission Checklist

- Flow names, types, purposes, inputs, outputs, decisions, record operations, and mappings documented.
- Full-canvas screenshots included for all four Agentforce-mapped Flows.
- Expected business errors and absence of technical Fault Connectors stated accurately.
- No Actual Results included; testing remains a separate submission artifact.